

Reporting Summary

Nature Portfolio wishes to improve the reproducibility of the work that we publish. This form provides structure for consistency and transparency in reporting. For further information on Nature Portfolio policies, see our [Editorial Policies](#) and the [Editorial Policy Checklist](#).

Statistics

For all statistical analyses, confirm that the following items are present in the figure legend, table legend, main text, or Methods section.

n/a	Confirmed
☒	☒ The exact sample size (<i>n</i>) for each experimental group/condition, given as a discrete number and unit of measurement
☒	☐ A statement on whether measurements were taken from distinct samples or whether the same sample was measured repeatedly
☒	☐ The statistical test(s) used AND whether they are one- or two-sided <i>Only common tests should be described solely by name; describe more complex techniques in the Methods section.</i>
☒	☐ A description of all covariates tested
☒	☐ A description of any assumptions or corrections, such as tests of normality and adjustment for multiple comparisons
☐	☒ A full description of the statistical parameters including central tendency (e.g. means) or other basic estimates (e.g. regression coefficient) AND variation (e.g. standard deviation) or associated estimates of uncertainty (e.g. confidence intervals)
☒	☐ For null hypothesis testing, the test statistic (e.g. <i>F</i> , <i>t</i> , <i>r</i>) with confidence intervals, effect sizes, degrees of freedom and <i>P</i> value noted <i>Give P values as exact values whenever suitable.</i>
☒	☐ For Bayesian analysis, information on the choice of priors and Markov chain Monte Carlo settings
☒	☐ For hierarchical and complex designs, identification of the appropriate level for tests and full reporting of outcomes
☒	☐ Estimates of effect sizes (e.g. Cohen's <i>d</i> , Pearson's <i>r</i>), indicating how they were calculated

Our web collection on [statistics for biologists](#) contains articles on many of the points above.

Software and code

Policy information about [availability of computer code](#)

Data collection	We used Micrisoft SQL Server to obtain data from the Mount Sinai Data Warehouse. Data from the MOVER dataset is available as csv files which were processed by Pandas==2.1.1
Data analysis	We write custom code for graph neural networks and loss functions for training. We use these libraries: torch==2.1.0 torch-geometric==2.4.0 torch-scatter==2.1.2 torch-sparse==0.6.18 torch-cluster==1.6.3 torch-spline-conv==1.2.2 torchvision==0.16.0 torchaudio==2.1.0 numpy==1.24.3 scipy==1.11.3 pandas==2.1.1 scikit-learn==1.3.0 networkx==3.1 PyTorch implementation and documentation are available at https://github.com/Nadkarni-Lab/InfEHR .

For manuscripts utilizing custom algorithms or software that are central to the research but not yet described in published literature, software must be made available to editors and reviewers. We strongly encourage code deposition in a community repository (e.g. GitHub). See the Nature Portfolio [guidelines for submitting code & software](#) for further information.

Data

Policy information about [availability of data](#)

All manuscripts must include a [data availability statement](#). This statement should provide the following information, where applicable:

- Accession codes, unique identifiers, or web links for publicly available datasets
- A description of any restrictions on data availability
- For clinical datasets or third party data, please ensure that the statement adheres to our [policy](#)

Mount Sinai Health System patient data used retrospectively for this project, and with institutional permission granted by the Institutional Review Board of the Icahn School of Medicine at Mount Sinai, comprise confidential patient records and are not publicly available due to privacy restrictions. Patient data from University of California, Irvine Medical Center are available through credentialed access as part of the MOVER dataset (<https://mover.ics.uci.edu/index.html>) with Institutional Review Board approval from University of California, Irvine School of Medicine and from University of California, Irvine main campus.

Processed data including model weights and embeddings are available at <https://github.com/Nadkarni-Lab/InfEHR>.

Research involving human participants, their data, or biological material

Policy information about studies with [human participants or human data](#). See also policy information about [sex, gender \(identity/presentation\), and sexual orientation](#) and [race, ethnicity and racism](#).

Reporting on sex and gender

We do not use sex or gender as inclusion or exclusion criterion in our studies or as features. Our studies are designed to reflect performance on the empirical distribution of cases in a hospital system. We wished to create models that would not explicitly use demographic information including sex and gender and so these information was not used in cohort selection or model development.

Reporting on race, ethnicity, or other socially relevant groupings

We did not use race, ethnicity, or any socially relevant groupings as inclusion or exclusion criterion or as features provided to models. We wished to create models that would focus on informative patterns in electronic health records without explicit use of demographic information including race, ethnicity, or any socially relevant groupings

Population characteristics

We did not use human research participants. Because we draw from the empirical sample of the patient population at the Mount Sinai Hospital System and data curated for the MOVER dataset (to our knowledge patients are included by merit of having a surgery at random) we reflect the empirical distribution of covariates as well.

Recruitment

We did not recruit any patients for this study

Ethics oversight

The Institutional Review Board at the Icahn School of Medicine at Mount Sinai approved this study.

Note that full information on the approval of the study protocol must also be provided in the manuscript.

Field-specific reporting

Please select the one below that is the best fit for your research. If you are not sure, read the appropriate sections before making your selection.

☒ Life sciences ☐ Behavioural & social sciences ☐ Ecological, evolutionary & environmental sciences

For a reference copy of the document with all sections, see [nature.com/documents/nr-reporting-summary-flat.pdf](https://www.nature.com/documents/nr-reporting-summary-flat.pdf)

Life sciences study design

All studies must disclose on these points even when the disclosure is negative.

Sample size

We included all possible cases that met inclusion criterion as defined in the manuscript. Sample size was not pre-determined other than as a byproduct of the inclusion criterion. We sought the maximum sample possible as it represents a test of scalability and it is also established that increasing examples is beneficial to deep learning models.

Data exclusions

We exclude cases when categorical missingness (for example no laboratory measurements or vitals signs information) was present or had otherwise failed to meet inclusion criterion. This was decided in advance. We exclude such cases because they either are not computable or represent instances of artifactual loss from database operations.

Replication

We used where possible fixed random seeds and could replicate results with these settings.

Randomization

In the culture-negative sepsis study we performed an internal validation split with stratification by birthweight as this is an independent risk factor for the condition of interest. Our data is otherwise split by institution of origin subject to the inclusion exclusion criterion in the manuscript.

Blinding

Our study is computational in nature and therefore blinding is not applicable.

Reporting for specific materials, systems and methods

We require information from authors about some types of materials, experimental systems and methods used in many studies. Here, indicate whether each material, system or method listed is relevant to your study. If you are not sure if a list item applies to your research, read the appropriate section before selecting a response.

Materials & experimental systems

n/a	Involved in the study
☒	☐ Antibodies
☒	☐ Eukaryotic cell lines
☒	☐ Palaeontology and archaeology
☒	☐ Animals and other organisms
☒	☐ Clinical data
☒	☐ Dual use research of concern
☒	☐ Plants

Methods

n/a	Involved in the study
☒	☐ ChIP-seq
☒	☐ Flow cytometry
☒	☐ MRI-based neuroimaging

Plants

Seed stocks

Report on the source of all seed stocks or other plant material used. If applicable, state the seed stock centre and catalogue number. If plant specimens were collected from the field, describe the collection location, date and sampling procedures.

Novel plant genotypes

Describe the methods by which all novel plant genotypes were produced. This includes those generated by transgenic approaches, gene editing, chemical/radiation-based mutagenesis and hybridization. For transgenic lines, describe the transformation method, the number of independent lines analyzed and the generation upon which experiments were performed. For gene-edited lines, describe the editor used, the endogenous sequence targeted for editing, the targeting guide RNA sequence (if applicable) and how the editor was applied.

Authentication

Describe any authentication procedures for each seed stock used or novel genotype generated. Describe any experiments used to assess the effect of a mutation and, where applicable, how potential secondary effects (e.g. second site T-DNA insertions, mosaicism, off-target gene editing) were examined.